

Sub-surface deformation of individual fingerprint ridges during tactile interactions

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By leveraging optical coherence tomography this study provides **important** insight into the deformation of human fingertip ridges when contacting raised features such as edges and contours. The study provides **compelling** evidence that such features tend to cause deformation and relative movement of what the authors term ridge flanks rather than bending of the ridges themselves.

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Abstract

The human fingertip can detect small tactile features with a spatial acuity roughly the width of a fingerprint ridge. However, how individual ridges deform under contact to support accurate and high-precision tactile feedback is currently unknown. The complex mechanical structure of the glabrous skin, composed of multiple layers and intricate morphology within which mechanoreceptors are embedded, makes this question challenging. Here, we used optical coherence tomography to image and track sub-surface deformations of hundreds of individual fingerprint ridges across ten participants and four individual contact events at high spatial resolution in vivo. We calculated strain patterns in both the stratum corneum and viable epidermis in response to a variety of passively-applied tactile stimuli, including static indentation, stick-to-slip events, sliding of a flat surface in different directions, and interaction with small tactile features, such as edges and grooves. We found that ridges could stretch, compress, and undergo considerable shearing orthogonal to the skin surface, but there was limited horizontal shear. Therefore, it appears that the primary components of ridge deformation and, potentially, neural responses are deformations of the ridge flanks and their relative movement, rather than overall bending of the ridges themselves. We conclude that the local distribution of mechanoreceptors across the ridges might be ideally suited to

extract the resulting strain gradients and that the fingertip skin may possess a higher mechanical spatial resolution than that of a single ridge.

Introduction

The human fingertip is a highly effective tactile sensing structure housing thousands of mechanoreceptors (Corniani and Saal, 2020 [↗](#)) that convert skin deformations into neural responses (Handler and Ginty, 2021 [↗](#)), which underlie our capacity for fine tactile discrimination and object manipulation (Lieber and Bensmaia, 2022 [↗](#); Johansson and Flanagan, 2009 [↗](#); Witney et al., 2004 [↗](#)). The ridged structure of the volar skin of the fingertip gives rise to fingerprints, which are thought to determine the spatial resolution at which we can resolve small tactile features. Ridge spacing correlates with perceptual spatial acuity (Peters et al., 2009 [↗](#)), and the size of receptive sub-fields of type-1 tactile afferents has been estimated in the sub-millimeter range, roughly matching the width of a single ridge (Jarocka et al., 2021 [↗](#); Sukumar et al., 2022 [↗](#)).

However, understanding how the mechanical properties of individual fingerprint ridges support high-precision tactile feedback presents significant challenges: any force applied to a papillary ridge on the skin's surface causes deformations within multiple skin layers with distinct mechanical properties and complex morphology (Dahiya and Gori, 2010 [↗](#)), before being transduced into neural activity by mechanoreceptors situated at the epidermis-dermis border, which are situated at specific landmarks relative to the limiting and intermediate ridge structures of the viable epidermis (see illustration in **Figure 1A** [↗](#)).

While numerous studies have investigated ridge deformations (Cauna, 1954 [↗](#); Gerling and Thomas, 2005 [↗](#), 2008 [↗](#); Willemet et al., 2021 [↗](#); Delhay et al., 2016 [↗](#)) and the mechanical properties of different skin layers (Pereira et al., 1991 [↗](#); Leyva-Mendivil et al., 2015 [↗](#), Leyva-Mendivil et al., 2017 [↗](#)), most of this work has focused on ex-vivo specimens, surface measurements, or computer simulations. Consequently, the mechanical response of the skin below its immediate surface remains largely unknown, leading to conflicting interpretations in the literature. For instance, it has been proposed that the papillary ridges are stiffer than the neighbouring grooves (Swensson et al., 1998 [↗](#)), which might imply that normal loading of the skin might not affect the ridges' profile appreciably. Conversely, other observations have suggested that the grooves are relatively stiff, allowing the papillary ridges to deform considerably (Cauna, 1954 [↗](#); Johansson and LaMotte, 1983 [↗](#)). However, the sub-surface consequences of this putative pliability during object contact or stick-to-slip transitions (see e.g. Delhay et al., 2016 [↗](#)) are unclear: the whole ridge structure might bend as proposed in Cauna's lever mechanism (Cauna, 1954 [↗](#)), but this view has proved controversial (see e.g. Gerling and Thomas, 2008 [↗](#)), with direct empirical evidence lacking.

From a mechanical standpoint, these conflicting interpretations raise the question of how the outermost two skin layers typically deform at the resolution of single papillary ridges, whether by tension, compression, or shear (see examples in **Figure 1B** [↗](#)). Additionally, such deformations might apply to individual papillary ridges and all their sub-surface structures equally, for example horizontal shearing that bends the papillary ridge in a certain direction, while levering its sub-surface aspects in the opposite direction. Conversely, individual parts of the ridge structure might deform differently. For example, the viable epidermis might deform to a different extent or in a different direction due to its lower stiffness and different morphology. Additionally, if there are indeed mechanical differences between papillary ridges and their neighbouring grooves at the level of the stratum corneum, this might result in differential movements of the two sides of each papillary ridge, here referred to as ridge flanks (see **Figure 1B-iv** [↗](#), right, for a potential example).

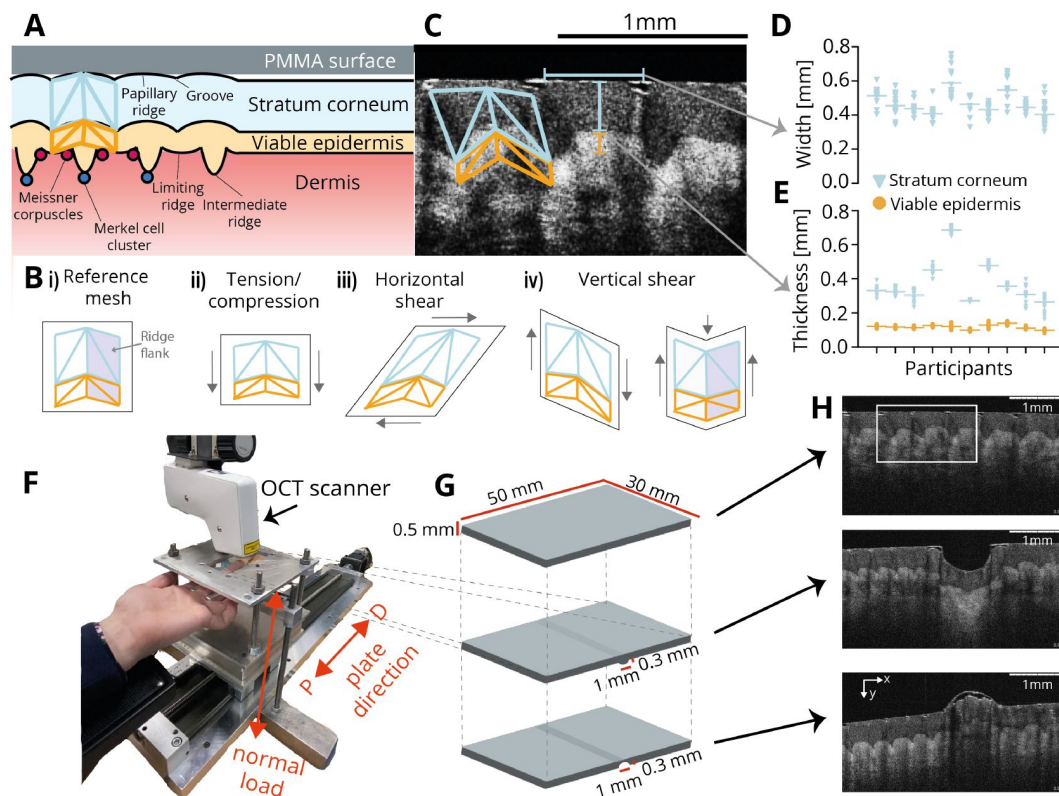


Figure 1.

Experimental setup.

A. Two major mechanically relevant skin layers constitute the epidermis on the human fingertip: the stratum corneum (blue shading) and the viable epidermis (orange). The papillary ridges making up the fingerprints at the skin surface extend to deeper layers, where morphological complexity increases. Interactions with surfaces (grey shading) take place at the outer boundary of the stratum corneum, while mechanoreceptors are located at the border between the epidermis and the dermis (red shading), at distinct landmarks associated with the ridge structure (dark red and blue circles). Identification of morphological landmarks related to the ridge structure allows the creation of a fine-grained mesh covering most of the sub-surface structure of a single papillary ridge (blue and orange overlaid meshes). **B.** Potential deformations of the tracked ridge structure, including the stratum corneum and the bulk of the viable epidermis, during tactile interactions, with arrows indicating the directions of relative deformation. i) Reference mesh (undeformed); ii) Tension or compression of the whole structure along or orthogonal to the surface axis of the skin; iii) horizontal (surface) shear, where the ridge structure tilts sideways; iv) vertical shear, where the ridge structure tilts along the axis orthogonal to the surface of the skin. These deformations might affect different parts separately, e.g. via shearing in different directions across both ridge flanks as shown on the far right (see darker shading to highlight a single ridge flank). **C.** Detail view of a single OCT frame showing ridged skin structure and clear boundary between the stratum corneum and viable epidermis. A mesh covering the stratum corneum and the upper part of the viable epidermis (without the intermediate ridge) is overlaid spanning a single papillary ridge. The border between the viable epidermis and dermis is less clearly delineated, but some deeper feature are resolved less well. **D.** Ridge widths ($n = 153$) across all participants. Each dot corresponds to a unique ridge with the participant mean indicated by the horizontal line. **E.** Thickness of stratum corneum (blue) and viable epidermis (yellow). Markers as in D. **F.** The experimental apparatus consisted of a finger holder to which the left middle finger of the participant was secured. A horizontal plate with a smaller inlaid transparent surface could be moved to indent the fingertip. A motorized linear stage moved the plate in the distal/proximal direction across the fingertip. An OCT scanner recorded images through the transparent surface, while forces were recorded using a 3-axis force sensor. **G.** Illustration of the three transparent PMMA surfaces: a flat surface, one embossed with a rounded edge, and a grooved one. **H.** Individual images recorded by the OCT scanner using the three surfaces shown in panel G. **Figure 1** [video 1. Recordings of sub-surface skin deformations.](#) Example recordings demonstrating stick-to-slip with the flat plate, and interactions with a groove and an edge, respectively. Each sequence shows first the raw data, and then the processed sequence with a single tracked ridge overlaid.

To empirically address these questions, we employed Optical Coherence Tomography (OCT) to precisely measure the sub-surface deformation of individual fingerprint ridges in response to a variety of mechanical events. Specifically, we focused on the stratum corneum and the bulk of the viable epidermis (excluding intermediate ridges), which could be robustly resolved and tracked by our setup. OCT has been used previously to investigate sub-surface skin properties, however, its application has been limited to static characterization of the skin morphology, such as measuring tissues' thickness and roughness (Maiti et al., 2020 [DOI](#); Ding et al., 2021 [DOI](#); Adabi et al., 2017 [DOI](#); Czekalla et al., 2019 [DOI](#); Lin et al., 2021 [DOI](#)). In the present study, we aimed to investigate the ways in which ridges are able to deform during dynamic contact events as well as the role played by different skin layers in order to better understand how the ridged structure of fingertip skin supports tactile sensing.

Results

Measuring skin deformation on a sub-ridge scale

We developed a setup for tactile stimulation of the fingertip, which involved lowering a custom-made transparent thin plate onto the participant's fixed fingertip at a set normal load and smoothly sliding the plate in both distal and proximal directions at a set speed (**Figure 1F, G** [DOI](#)). Concurrently, we used an OCT scanner to capture images of the skin's surface and sub-surface morphology. The scanner obtained 4 mm wide slices with 4.5 μm lateral and 5 μm axial resolution at a sampling frequency of 10 Hz. The images were taken by scanning through the transparent plate along the proximal-distal axis of the finger. Both the stratum corneum and viable epidermis were clearly distinguishable from each other (**Figure 1C, H** [DOI](#)).

After acquiring and pre-processing the OCT images, we used semi-automatic methods to track sub-surface landmarks associated with individual ridges across different video frames (see Methods, Figure 1—video 1). The resulting triangular mesh consisted of eight facets per ridge, distributed between the stratum corneum and viable epidermis, as well as the two flanks of the ridge (**Figure 1B** [DOI](#)). Importantly, the mesh was fine-grained enough to identify canonical sub-surface ridge deformation patterns in response to various tactile events. The mesh also extended to and partially covered the dermis-epidermis border, where low-threshold mechanoreceptors are located (**Figure 1A** [DOI](#)). We measured an average ridge width of 0.47 mm across participants (**Figure 1D** [DOI](#)), consistent with previous studies (Moore, 1989 [DOI](#); Ohler and Cummins, 1942 [DOI](#)). Average skin layer thickness was 0.38 mm for the stratum corneum and 0.12 mm for the viable epidermis across our dataset (**Figure 1E** [DOI](#)), again in agreement with previous studies using both in vivo imaging and ex vivo histology (Fruhstorfer et al., 2000 [DOI](#); Lintzeri et al., 2022 [DOI](#); Maiti et al., 2020 [DOI](#)). We gathered a large and varied dataset to be able to make robust claims: across four protocols and ten participants, we identified a total of 393 unique ridge slices, which were tracked for 34 s on average and more than a minute in some cases, supported by hundreds of thousands of individually tracked landmarks.

Static normal load

To investigate the response of individual ridges to static normal loads, we applied a flat stimulus to the fingertip, incrementally increasing the load from initial contact in steps of 0.5 N until reaching a maximum of 3.5 N (see examples in **Figure 2A, B** [DOI](#)). Deformation was measured relative to the unloaded ridge's stereotypical configuration, and the principal tensile and compressive components of the strain experienced by each mesh facet were obtained (see Methods).

Under load, the central part of individual ridges depressed, while the flanks stayed in place (vertical shear), which led to a flattening of the ridge structure and which was consistently observed across participants (see examples in **Figure 2C** [DOI](#)). Ridges could also display horizontal shear and bend proximally or distally depending on their unloaded configuration and sometimes

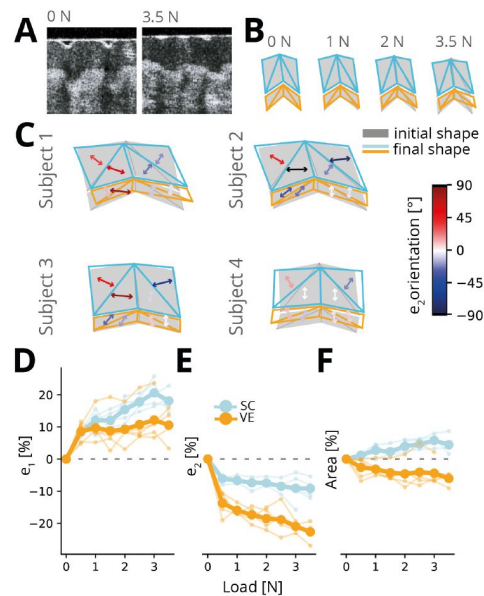


Figure 2.

Ridge deformation under normal load.

A. Example frames showing a single ridge unloaded (0 N; left) and maximally loaded (3.5 N; right). **B.** Examples of ridge deformation under static normal load for different load conditions. Shown is the unloaded ridge outline (grey shaded regions) with the deformed ridge mesh superimposed (blue and orange), under 1 N, 2 N, and 3.5 N loading. **C.** Examples of ridge deformation under static normal load for different participants. Shown is the unloaded ridge outline (grey shaded regions) with the ridge mesh under 3.5 N loading superimposed. Arrows indicate the principal compressive axis for each facet, with colors indicating different orientations relative to vertical. In the lower panels, the magnitude of principal tensile (e_1 , **D.**) and compressive (e_2 , **E.**) strains as well as area change (e_a , **F.**) as a function of normal load. Thin lines denote individual participants, and thick lines show the average. Data from five participants with seven individual ridges tracked each.

stretch horizontally along the axis of the skin surface. Both tensile and compressive strains increased monotonically with normal load, with rapid changes and, therefore, larger deformations at initial contact followed by slower and mostly linear changes from 0.5 N onwards (**Figure 2D, E** [↗](#)). Across all facets in a given layer, tensile strains reached around 15% in the stratum corneum and 10% in the viable epidermis, while compressive strains were above 20% in the viable epidermis and steadily increased with the load but remained below 10% in the stratum corneum. We noted a small increase in the area of the stratum corneum, which was possibly an artifact due to the imperfect fit of the mesh to the ridge's curvature (but see Discussion for an alternative explanation). In contrast, we found a notable decrease in the area of the viable epidermis (**Figure 2F** [↗](#)). Our results, therefore, indicate that both the stratum corneum and the viable epidermis undergo deformation under static load, manifested predominantly as a flattening of the papillary ridge and its sub-surface structure, but that compression is higher in the viable epidermis.

Lateral sliding and stick-to-slip

After characterizing the deformation resulting from a static normal load, we investigated sliding interactions between the fingertip and the flat surface. The flat surface was first lowered onto the fingertip until stable contact was established (with a normal load below 0.5 N), then displaced at a constant speed of 0.8 mm/s by 7.6 mm in the distal or proximal direction for a total of 8 movements. In accordance with previous findings, we observed two distinct phases of relative movement between the skin and the plate: stick, where the ridges adhered to the plate and were dragged along with it, and slip, where the ridges stopped moving, and the plate slipped over them. The clear distinction between the stick and slip phases allowed us to classify each phase based on the average horizontal velocity over all tracked features (see **Figure 3A** [↗](#) for an example from a single participant).

Previous studies have shown that stick-to-slip events elicit strain waves on the surface of the fingertip skin (Delhay et al., 2021 [↗](#); Willemet et al., 2022 [↗](#); du Bois de Dunilac et al., 2023 [↗](#)), but their effects on the sub-surface structure are unclear. Additionally, reversing the plate's movement direction changes the tangential force applied to the skin and induces shear deformations within the tissue, but it is unclear whether these are absorbed in surface skin layers or in deeper tissues. To quantify the deformation that occurred between the transitions from stick to slip and between different movement directions of the plate, respectively, we calculated a stereotypical mesh covering a single papillary ridge by averaging the meshes of all tracked ridges ($n = 140$ in total) across image frames within the same phase for each participant (**Figure 3B, C** [↗](#), see Methods). We then calculated the principal tensile and compressive strain components experienced by each mesh facet on the transition from one phase to the next.

Examining the ridge deformation changes between movement phases revealed large variability across participants, both in the magnitude and the orientation of the deformations (see individual examples in **Figure 3D** [↗](#) and all participants in **Figure 3—Figure Supplement 1** [↗](#)). This variability is likely explained by the fact that, during stick-to-slip transitions, some parts of the fingertip experience tension, while others experience compression (Delhay et al., 2016 [↗](#)), and therefore measurements depend on the precise location. Additionally, the mechanical properties of each individual fingertip influence its own mechanical response, and crucially, the timing of the stick-to-slip transition during plate transit (see Methods), leading to varied mechanical environments. Nonetheless, several consistent findings emerged from the data.

First, we found that, across participants, strain magnitudes (quantified as maximal shear strain, see Methods) were consistently larger by about a third during the stick-to-slip transitions compared to the plate reversal transition (**Figure 3E** [↗](#)) for both the stratum corneum ($p < 0.01$, paired Wilcoxon signed-rank test across all facets, movement directions, and participants) and viable epidermis ($p < 0.01$). This was also true for individual participants, with strain largest

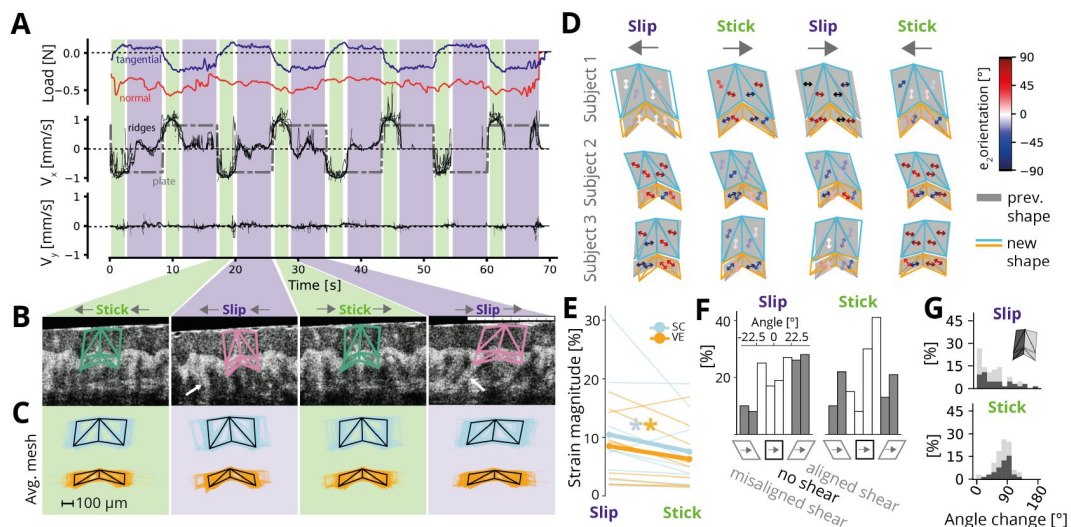


Figure 3.

Ridge deformations during sliding.

A. Measurements during repeated movements of the flat plate along the distal-proximal axis for a single participant. Top: Tangential (purple) and normal (red) load as a function of time. The normal load was set at 0.5 N at the start of the trial by adjusting the indentation and then was not further controlled. Tangential load alternates between positive and negative values depending on the movement direction of the plate. Middle: Horizontal velocity of the plate (dash-dotted grey line) and average velocity of each tracked fingerprint ridge (thin black lines) during all eight transitions of the flat plate. Two phases are evident: when the ridge is moving along with the plate (stick, indicated by green shading) and when the ridge is stationary, but the plate is moving (slip: indicated by pink shading). Bottom: Vertical velocity of all tracked ridges along the normal axis, which is close to zero. **B.** Example frames showing a single-tracked ridge for the stick and slip phases of the two movement directions. White arrows point to presumed collagen fiber bundles anchoring the skin to the bone. **C.** Average ridge meshes (black lines) during each of the four phases calculated over all tracked ridges and all time points assigned to each phase for the same participant as in A, B. Colored lines indicate individual sample meshes ($n = 3773$; blue: stratum corneum, yellow: viable epidermis). **D.** Examples showing ridge deformation across the four phases for three participants. Shown are the ridge outline for the previous phase (grey shaded regions) with the ridge mesh for the current mesh superimposed. Arrows indicate the principal compressive axis for each facet, with colors indicating different orientations relative to vertical. **E.** Average magnitude of maximal shear strains as a proxy for overall deformation in the stick-to-slip transition compared to the movement reversal. Thin lines denote individual participants (averaged over facets and movement directions), and thick lines show the grand average. Asterisks denote statistically significant differences (paired Wilcoxon tests). Strains are about a third higher during stick-to-slip transitions than during movement reversals. **F.** Histograms of average principal strain angles for all mesh facets and participants in stick-to-slip transitions (left) and movement reversals (right). White bars denote angles that are within 22.5° of the coordinate axes (horizontal or vertical) and therefore denote tension or compression without considerable shear. Grey bars denote angles within 22.5° of the diagonal and therefore denote horizontal shear. Positive angles denote shear acting in the same direction as the plate movement, while negative angles denote shear in the direction opposite to the plate movement. **G.** Change in principal strain angles when transitioning to slip (top) or stick (bottom) phases, separated by ridge flank (light and dark grey). Movement reversals cause a 90-degree shift in the strain angle, while stick-to-slip transitions cause little change, with no differences between ridge flanks evident. **Figure 3—figure supplement 1** [↗](#). Ridge deformations for all participants during sliding of the flat plate.

during stick-to-slip transitions for eight out of nine participants. Thus, ridge deformations are most pronounced during stick-to-slip transitions rather than in response to different movement directions of the surface.

Second, while the orientation of the principal strain axes varied between participants, they were relatively consistent within participants in individual phases: the average angular standard deviation was 26 degrees when considering all facets of the ridge within a given phase and, subsequently, averaging over phases and participants. Consequently, strains tended to act in a consistent direction along the entire extent of the ridge (see also examples in [Figure 3D](#)). Notably, the principal strain directions exhibited a greater alignment with the cardinal axes (0 and ± 90 degrees) rather than diagonal axes (± 45 degrees) for the majority of participants across most phases (averaging 60% overall, see [Figure 3F](#)). This alignment implies a predominance of tension and compression in the ridge rather than horizontal shear. When shear did manifest, it was more pronounced during stick-to-slip transitions compared to direction changes, and generally aligned with the direction of plate movement, although this alignment was not always observed. We did not find any significant differences in the distribution of principal angles between the stratum corneum and the viable epidermis for either stick-to-slip transitions or movement reversals ($p > 0.37$ for both based on two-sample Kolmogorov-Smirnov tests across all facets, directions, and participants).

The low amount of shear was surprising, as during stick phases the skin was typically dragged along with the surface for several millimeters (mean: 3.4 mm, range: 0.9-6.3 mm). Assuming no motion of the bone, and a distance of 5-10 mm between the lightly loaded skin surface and the bone (Birznieks et al., 2001) yields large expected shear strains of 45% on average (range: 9-126%), clearly much in excess of what we observed. Thus, it is possible that shearing is sustained by deeper tissues, an effect that could be tested in future studies by directly tracking the angle and orientation of collagen fiber bundles anchoring the epidermis to deeper tissues (see highlighted examples in [Figure 3B](#)).

Finally, when switching movement direction, which caused the skin to stick to the surface again, we observed a 90-degree rotation of the principal axis of deformation (see [Figure 3G](#), bottom panel). In contrast, there was relatively little change in the angle when transitioning to slip ([Figure 3G](#), top panel). Thus, changing direction caused a reversal in the tension and compression axes but relatively little deformation in itself; this deformation then occurred during the transition to slip. The observed changes in orientation occurred consistently for both ridge flanks and therefore applied uniformly to the whole ridge.

In summary, our findings indicate that the primary deformations occurring during contact with a flat plate predominantly involve tension and compression of individual ridges along the skin's surface. Horizontal shear is only occasionally present and does not consistently align with the direction of movement. When shear is observed, it is more prominent during stick-to-slip transitions rather than during reversals of the plate movement.

Ridge interactions with small tactile features

Finally, we investigated ridge deformation during contact with small tactile features that were close in size to that of a single ridge itself (1 mm in width and 0.3 mm in height/depth). For this experiment, we used the plates with the edge and groove, respectively, keeping all other stimulation parameters the same. We identified five consecutive phases for each tracked ridge based on its position relative to the edge or groove: 1) sliding under the flat part of the surface, 2) approach of the feature, 3) located centrally under the feature, 4) withdrawal from the feature, and 5) sliding under the flat part of the surface (see illustrations in top rows of [Figure 4A, D](#)). We again quantified the deformation of the average ridge during all these phases (see Methods), with the strains calculated relative to the initial mesh before contact with the feature was made (phase 1).

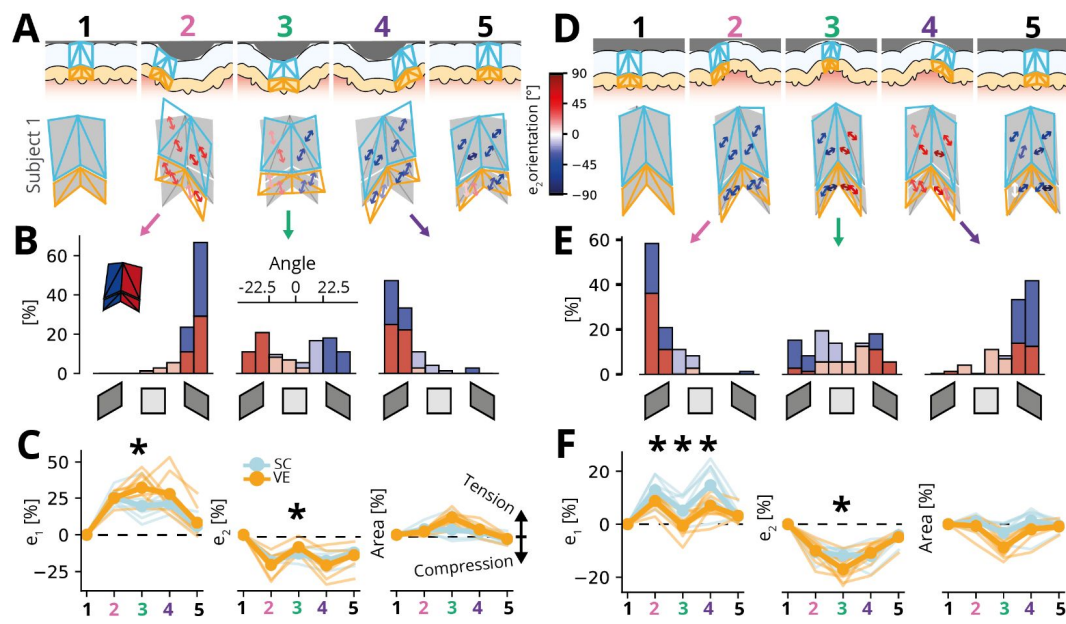


Figure 4.

Ridge deformations and skin strains during transit of small tactile features.

A. Top row: Illustration of identified movement phases with the edge feature in different locations relative to the tracked fingerprint ridge. Bottom row: Average ridge deformation and associated principal compressive strain orientation for a single participant. Note that all strains are calculated with respect to the mesh shown in the left-most column, which represents the ridge during full slip before the interaction with the tactile feature. **B.** Histograms of principal strain orientations across all ridge facets and participants for the approach (2), central (3), and withdrawal (4) phases. Red bars denote distal ridge flanks and blue bars denote proximal ones. Darker shading denotes orientations close to diagonal, indicating shear, while lighter shadings denote angles aligned vertically or horizontally and therefore denote no shear. **C.** Average magnitude of principal tensile (e_1 , left panel) and compressive (e_2 , middle panel) strains as well as area change (right panel) relative to the stereotypical ridge under full sliding (phase 1). Thin lines denote individual participants ($n = 9$), and thick lines show the average. Asterisks denote statistically significant differences between the stratum corneum and the viable epidermis (Bonferroni corrected paired Wilcoxon tests). Ridges are most deformed when directly under the edge feature when strains are higher in the viable epidermis than the stratum corneum. While the stratum corneum is incompressible, the area of the viable epidermis expands when directly under the ridge. **D-F.** Same as in A-C, but for a small groove transitioning over the ridge. Note that the findings broadly reflect those obtained with the edge feature, but with the direction of compressive and tensile strains reversed. **Figure 4—figure supplement 1** [↗](#). Ridge deformations for all participants during transit of the edge feature. **Figure 4—figure supplement 2** [↗](#). Ridge deformations for all participants during transit of the grooved feature.

Ridge deformations were remarkably similar across participants but differed clearly across movement phases (see ridge deformations for a single participant in [Figure 4A, D](#) bottom rows and all participants in [Figure 4—Figure Supplement 1](#) and [Figure 4—Figure Supplement 2](#)). There was minimal evidence supporting the notion of a rigid ridge maintaining its shape, and instead, the central part of the ridge deformed considerably. Horizontal shear was relatively modest, with generally little bending of the ridge observed. Instead, our findings highlighted that the primary mechanism through which the ridge conformed to the shape of a feature involved vertical shearing of the whole ridge or the two ridge flanks against each other, as the skin moved vertically to conform to the feature. Consequently, during the transit of the edge, the full ridge initially sheared orthogonally to the skin's surface with both the strain orientations of both flanks aligned ([Figure 4B](#), left panel). When the edge was directly above the ridge, both flanks sheared against each other, leading to opposite strain orientations across the two flanks ([Figure 4B](#), middle panel) before the ridge returned to its original configuration during the withdrawal of the edge ([Figure 4B](#), right panel). This pattern was mirrored, but with strain orientations reversed, during the transit of the groove ([Figure 4E](#)).

Strains increased and reached their maximum with respect to the initial configuration when the ridge was located directly under the feature and then decreased again as the ridge reverted to its original shape. For the edge, both strain components differed between the viable epidermis compared to the stratum corneum when the ridge was located directly below the feature ($p < 0.01$ in Bonferroni corrected paired Wilcoxon tests across all facets and participants), resulting in an expansion of the area of the viable epidermis, as it was stretched around the edge feature, while the mostly incompressible stratum corneum retained its area ([Figure 4C](#)). Conversely, while interacting with the groove, a ridge experienced high compressive strains, which again were larger in the viable epidermis compared to the stratum corneum, and as a result, the area covered by the viable epidermis contracted, while the stratum corneum retained its area ([Figure 4F](#)).

Discussion

This study measured the sub-surface deformation of individual papillary ridges in the fingertip skin during tactile interactions in vivo. The dataset included hundreds of individual ridges, whose deformations were tracked using high-resolution meshes during static normal load, lateral stick, and slip using a flat surface, as well as during contact with small tactile features. We found that individual ridges tended to flatten considerably at relatively low forces of 0.5 N, with higher forces increasing deformations only moderately. Lateral sliding elicited strong tension and compression, often aligned with the surface orientation of the skin, which was especially strong during stick-to-slip transitions. Contact with edges and grooves yielded strong vertical shear, with ridge flanks moving relative to each other. We observed horizontal shear during static contact and sometimes when sliding the flat surface, however, it was limited in magnitude and was present only for some participants.

Canonical ridge deformations during tactile interactions

One of the main objectives of our study was to investigate the primary deformation patterns exhibited by individual ridges during tactile interactions. Previous studies combining mechanical measurements at the ridge surface and histology (Cauna, 1954) or analyzing the spatial keratin expression patterns within single ridges (Swensson et al., 1998) have suggested the presence of alternating softer and stiffer regions within the ridges. These findings implied a mechanically constrained set of canonical deformations, yet the specific nature of these deformations has been a subject of debate, lacking direct empirical evidence.

In the present study, we have provided conclusive evidence that ridge peaks are not rigid and, instead, individual ridges undergo considerable flattening during contact. This finding challenges previous speculation that the grooves between the papillary ridges act as ‘hinges’ (Swensson et al., 1998 [↗](#)). Instead, our results suggest that the fingerprint predominantly conforms to an external object through vertical shearing. Therefore, we propose that the primary mechanism through which a ridge conforms to an object might involve the relative movement and shearing of the ridge flanks, rather than relying on the grooves as articulated joints.

In contrast to the prominent vertical shearing observed, our study revealed minimal and inconsistent horizontal shear. We propose that the flattening of the ridges during contact may prevent further bending of the ridges, thus limiting the occurrence of horizontal shear. The absence of horizontal shear during the stick phase of lateral sliding on the flat surface was particularly unexpected. During these periods, the skin re-adhered to the plate and was subsequently dragged along, often over a distance of several millimeters. It is possible that the majority of the surface movement of the skin was absorbed by deeper tissues rather than the surface layers of the skin imaged in the present study. If that is the case, recent modeling work has suggested that tissue deformations are highly dependent on the orientation of collagen fibers in these tissues (Duprez et al., 2024 [↗](#)), which might be amenable to tracking in future OCT work to test this idea directly. Additionally, previous work investigating tactile afferent responses to tangential skin movements has reported strong activation of SA-2 receptors, thought to measure skin stretch mainly in deeper tissues (Saal et al., 2025 [↗](#)), providing further indirect evidence.

Implications for neural coding of touch

Ridge deformations are ultimately transduced into neural activity by mechanoreceptors, and our findings provide new insights into this process.

First, the magnitude of sub-surface deformation measured in our study correlated with neural response strengths observed in other studies. In trials with the flat surface, we measured larger deformations during stick-to-slip transitions than during reversals of the plate’s movement direction. Analogously, a previous study recorded neural responses during sliding stimulation of the fingertip and reported that rapidly adapting type I afferents (RA-I) showed little to no response during the onset and offset of the tangential load but strongly encoded the stick-to-slip transition (Delhay et al., 2021 [↗](#)), suggesting that mechanoreceptors directly encode the local amount of deformation. Our findings also extend these prior results by demonstrating that the large surface strains measured during the transition from stick to slip are mirrored in tissues below the surface and specifically at the depth at which type-1 mechanoreceptors are located. Altogether these results provide a physiological explanation for the widely observed human capacity of fine-tuned grasping with quick adjustments of the grip force to friction changes (Johansson and Westling, 1984 [↗](#); Cadoret and Smith, 1996 [↗](#); Delhay et al., 2024 [↗](#)).

Second, our finding that contact with small tactile features elicits shearing of both ridge flanks against each other suggests that the mechanical resolution of the fingerprint skin is on the order of half the width of an individual papillary ridge. In principle, such fine-grained and local differences in strain might be detectable by mechanoreceptors. Indeed, Meissner corpuscles innervate each flank of a fingerprint’s ridge, and prior work has shown that separate fibers can innervate the two flanks of the same ridge (Nolano et al., 2003 [↗](#)). Previous psychophysical work also appears to support this idea because it has estimated the spatial resolution of static touch at around 200 μm , thus approximately half the width of a single ridge of the fingerprint (Hollins and Risner, 2000 [↗](#); Bens-maia and Hollins, 2003 [↗](#)). However, it should be pointed out that such estimates are notoriously difficult to obtain reliably as they depend on both the precise stimulus and the mode of contact and that this limit has traditionally been associated with the responses of Merkel cells rather than Meissner corpuscles. Previous neural recordings measured the spatial acuity of the receptive subfields of RA1 and SA1 afferents, which were estimated to lie in the submillimeter range, roughly matching the width of a single ridge (Jarocka et al., 2021 [↗](#); Sukumar et al., 2022 [↗](#)).

Our results also suggest that complementary features (edges and grooves, respectively) generate symmetrical deformation. Recording responses from the same tactile afferent to such complementary features might help to disentangle what aspects of shearing the afferent is sensitive to. Therefore further electrophysiological and psychophysical work is needed to determine whether the sub-ridge mechanical resolution of the fingerprint skin does indeed translate into corresponding neural and perceptual acuity.

Limitations and future work

While our study was based on a large dataset, which contained a variety of tactile interactions, there are a number of limitations. First, while we could confidently track landmarks associated with the stratum corneum, we could not reliably identify intermediate ridges in the viable epidermis, though they were visible in some of the frames, limiting the depth of the fitted mesh. We hypothesize that the additional depth of these ridges combined with their slender morphology might have degraded the signal. 3D OCT imaging (see below) might help to resolve these features in future work and settle open questions regarding their precise morphology. Intermediate ridges are of importance because Merkel cell clusters are located at their tip, and they are therefore likely important for mechanotransduction. Indeed, intermediate ridges feature in the so-called ‘lever hypothesis’ of tactile transduction (Cauna, 1954 [\[1\]](#)), which has been called into question (Gerling and Thomas, 2005 [\[2\]](#), 2008 [\[3\]](#)), though direct empirical evidence is currently lacking. Ideally, future studies would improve imaging resolution and clarity to visualize morphological structures in more detail and at greater depth. We employed a state-of-the-art, clinical OCT scanner, routinely used in dermatological clinical assessments. Data from a recently developed experimental scanner demonstrated the possibility of resolving smaller structures in the tissues, even as small as Meissner corpuscles, by applying a technique called speckle-modulating OCT (Liba et al., 2017 [\[4\]](#)). Coupling such advanced imaging techniques with our experimental set-up and analysis pipeline would further advance the understanding of sub-surface skin properties and deformation mechanisms and allow more direct investigation of neural mechanisms. Such advanced imaging could also be combined with electrophysiological recordings from afferent fibers using microneurography (see Delhay et al., 2021 [\[5\]](#), for an example).

Second, while the edges and grooves of our feature plates were only marginally larger than a single papillary ridge and mimic those employed in other studies (Jarocka et al., 2021 [\[6\]](#)), even smaller stimuli would be capable of delivering much more localized forces (Johansson and LaMotte, 1983 [\[7\]](#); Kao et al., 2022 [\[8\]](#)). In the present study, we were limited by the resolution of the 3D printing technique employed to manufacture the transparent plates. Additionally, sharp corners elicit optical artifacts in the obtained images, which can occlude landmarks.

Third, we obtained two-dimensional slices oriented orthogonal to the orientation of the fingerprints, but the skin is a three-dimensional structure. Some of our tactile interactions might have caused skin deformations out-of-plane that were thus not measurable. For example, the slight increase in thickness of the stratum corneum under normal load might be explained as a measurement artifact due to the coarse nature of the mesh fitted, but could alternatively reflect tissue from out-of-plane regions pushing into the imaged space. Indeed, recent surface measurements of the skin’s behaviour during initial object contact have reported compression of the skin in the plane parallel to its surface (Doumont et al., 2025 [\[9\]](#)), which would result in increasing thickness, assuming that the stratum corneum is incompressible. Future studies could consider creating three-dimensional reconstructions of the fingerprint structure to study such effects.

Fourth, in the present study, we focused on young, healthy participants, who were tested in a temperature and humidity-controlled environment. However, several studies have shown that morphological and mechanical skin properties can greatly vary with physiological and physical factors such as age (Jobanputra et al., 2020 [\[10\]](#); Püllen et al., 2021 [\[11\]](#); Skedung et al., 2018 [\[12\]](#)), sex (Luebberding et al., 2014 [\[13\]](#); Diridollou et al., 2000 [\[14\]](#)), skin hydration (Tomlinson et al., 2011 [\[15\]](#)),

moisturization (Dione et al., 2023 [↗](#)), as well as ambient temperature and humidity levels (Klaassen et al., 2016 [↗](#)). All these factors might therefore also influence how individual ridges deform and present opportunities for future studies.

Finally, while we focused on the fingertip only, many other skin regions present interesting mechanical challenges waiting to be explored. The general ridged structure observed on the fingertip is common to all glabrous skin, but the local ridge mechanics might still differ: glabrous skin on the foot sole exhibits some morphological differences in order to support large weights that might well influence its mechanical response (Boyle et al., 2019 [↗](#)). For example, the morphology of transverse ridges (running orthogonal to and connecting limiting with intermediate ridges) differs across regions on the foot sole (Nagashima and Tsuchida, 2011 [↗](#)) and very likely from the hand (Yamada et al., 1996 [↗](#)). Our method should be directly applicable to study deformations of these ridges, though three-dimensional observations might be needed to resolve some of the open questions. Hairy skin, in contrast, differs from glabrous skin in that the stratum corneum is much thinner. It also lacks the clearly organised ridge structure, but exhibits more loosely oriented skin folds instead, which very likely also serve a mechanical function (Leyva-Mendivil et al., 2015 [↗](#)) and in principle are amenable to study using OCT.

Methods and Materials

Participants

Ten healthy participants (3 males, average age 20.5 years) participated in the experiment after providing informed consent. Due to low imaging quality for data from one participant, which precluded consistent tracking, all sliding experiments included data from nine participants; the static protocol was run on a subset of five participants. The experiment was performed on the subjects' left middle finger. Before any trial, the finger was thoroughly cleaned with water and dried carefully, and a thin layer of petroleum jelly was applied to moisturize the skin; this helped prevent repeated stick-slip events while sliding the surface. Environmental conditions during the test were controlled with 20° C room temperature and 50% relative humidity. The study protocol received ethical approval from The University of Sheffield Research Ethics Committee (ethics number 039144). All experiments took place in the Skin Barrier Research Facility, operated by Sheffield Dermatology Research at the Royal Hallamshire Hospital in Sheffield, UK.

Experimental set-up

The experimental set-up for the present study was adapted from the design of an earlier study (Lee et al., 2019 [↗](#)). Transparent plates were secured by a support rig fixed on a linear stage that could move in both distal and proximal directions with respect to the participant's fingertip, allowing the plate to slide against the finger. The fingertip was glued onto a finger holder to maintain its position during the acquisitions. The support rig was mounted on a force plate (HE6X6-10, Advanced Mechanical Technology, Inc.) to measure the load along the axes parallel and perpendicular to the direction of the movement and along the axis normal to the plate surface. The plates were made from polymethyl methacrylate (PMMA) and were 30×40×0.5 mm in size (produced by Shape Technology S.r.l., Casale Monferrato, Italy). One plate was flat, while the two others had an embossed oriented half-circular edge (1 mm base diameter and 0.4 mm height) or an engraved oriented groove (1 mm based diameter and 0.3 mm depth) traversing the middle of the surface (see **Figure 1F** [↗](#)).

Prior to each trial, the plate specimens were cleaned with deionized water and dried thoroughly with paper towels. The participant's finger was secured in a finger holder, which was positioned in such a way that the flat part of the fingertip distal to the whorl made initial contact with the plate as it was lowered onto the fingertip. The scanner was positioned such that its scan path aligned with the distal-proximal axis of the plate, targeting the centre line of the fingerpad so that the

fingerprint ridges were oriented orthogonally to the line scan. In the static loading experiment, the plate was initially positioned in contact with the participant's fingertip without exerting any pressure (~ 0 N). Afterward, the load was manually incrementally increased in 0.5 N steps, reaching a maximum of 3.5 N. At each step, 20 frames were recorded. In the sliding experiments, before commencing the imaging acquisition in each trial, the plate was manually lowered onto the participant's fingertip until a normal load of 0.2 N was reached. During each trial, the plate was moved 7.6 mm in each direction (proximal to distal and vice versa) four times at a constant speed of 0.8 mm/s. For these experiments, imaging focused on the central flat part of the contact area, such that all fingerprint ridges visible in the imaged region were in contact with the plate throughout the trial.

Image acquisition and pre-processing

For image acquisition, we utilized the Vivosight OCT system (Michelson Diagnostics, Kent), which is clinically approved and features a Fourier domain with a 20 kHz swept-source laser at 1300 nm center wavelength. The image capture rate was 10 frames per second, and the dimension of each image was 895×483 pixels with 256 gray levels with a $4.5 \mu\text{m}$ lateral and $5 \mu\text{m}$ axial resolution.

Following the acquisition, we preprocessed the images using the cv2 library in Python. Firstly, we normalized the brightness and increased the contrast by applying histogram equalization. Then, to further enhance the contrast and adjust the image saturation, we performed a gamma transformation ($\gamma = 5$). Lastly, we removed noise while preserving sharp edges by applying a bilateral filter (diameter of pixel neighborhood: 5, sigmaColor: 110, sigmaSpace: 190).

Tracking of individual ridge deformation

In each frame of the experiment, the top and bottom of each ridge within the field of view were semi-automatically tracked at three levels: the surface, the border between the stratum corneum and viable epidermis, and the junction between the dermis and epidermis. As imaging slices were 4 mm wide, often 8-9 ridges were visible concurrently. For each set of frames obtained during a single trial, a portion of the frames, ranging between 25-50% of the total, were manually annotated using the Python polygonal annotation library Labelme (Wada et al., 2021 [DOI](#)). The annotated frames were then used to train a DeepLabCut model to automatically estimate the position of each tracked point in the remaining frames (Nath et al., 2019 [DOI](#)). The accuracy of the models, on average, was 2.96 pixels on the training set and 14.64 pixels on the test set. We then manually reviewed the automatic annotations to correct any inaccuracies that might have arisen from automatic tracking. To measure the deformation of individual sub-surface ridges, the tracked points were used to create a triangular mesh for each ridge in the field of view. The mesh consisted of 8 facets per ridge, divided between the stratum corneum and viable epidermis and between the proximal and distal flanks of each ridge.

Movement phase classification

To study the skin movement during the transits of the plate, we analyzed the displacement of the tracked points from frame to frame. We calculated the horizontal and vertical components of the skin velocity, v_x and v_y , respectively, by measuring the displacement along the x-axis and y-axis. To determine the velocity components for each ridge, we averaged the velocities of the nine tracked points belonging to that ridge, resulting in v_x^{ridge} and v_y^{ridge} .

For the flat plate trials, four phases were identified by analyzing the horizontal component of the velocity v_x^{ridge} :

1. Sticking proximally ($v_x^{\text{plate}} < 0$, $v_x^{\text{ridge}} < 0$): the plate is moving in distal to proximal direction, and the skin is sticking to the plate.

2. Slipping proximally ($v_x^{plate} < 0$, $v_x^{ridge} = 0$): the plate is moving in distal to proximal direction, and it is slipping over the stationary skin.
3. Sticking distally ($v_x^{plate} > 0$, $v_x^{ridge} > 0$): the plate is moving in proximal to distal direction, and the skin is sticking to the plate.
4. Slipping distally ($v_x^{plate} > 0$, $v_x^{ridge} = 0$): the plate is moving in proximal to distal direction, and it is slipping over the stationary skin.

To identify the transition between sticking and slipping phases, we took the positive and negative extrema of the first derivative of v_x^{ridge} as markers. Each trial consisted of four plate transits in each direction, resulting in a total of 16 successive phases. We selected the central n_{phase} frames for each phase to ensure an equal number of frames for each occurrence of a single phase during the subsequent mesh averaging. We chose n_{phase} as the minimum duration of each phase among its four repetitions. The time spent in either phase varied markedly across participants (range: 11.4–82.7%, mean: 44.7%). These differences are partly attributable to differences in skin mechanics across participants (e.g., moisture level) but also depend on the precise location of the imaging site on the fingertip, as previous research has shown that stable contact is lost progressively over time across the whole fingertip during the transition from stick to slip (Delhaye et al., 2014 [DOI](#)).

For trials using the feature plates, all frames where $v_x^{ridge} \neq 0$ were discarded to retain only data where the skin was fully slipping under the plate. Then, by examining the vertical component of the skin speed v_y^{ridge} , four phases were classified for every ridge mesh separately:

1. full slipping ($v_y^{ridge} = 0$): before and after approaching the feature
2. approaching the feature ($v_y^{ridge} > 0$ for the groove plate and $v_y^{ridge} < 0$ for the edge plate)
3. moving away from the feature ($v_y^{ridge} < 0$ for the groove plate and $v_y^{ridge} > 0$ for the edge plate)
4. under the plate feature (frames in between the two previous phases, $v_y^{ridge} = 0$.)

In each trial, there were a total of 40 successive phases, as each of the 5 phases identified occurred in each of the 8 transits of the plate. Again, the central n_{phase} frames were selected for each phase in order to consider the subsequent mesh averaging an equal number of frames in each occurrence of a single phase.

Strain computation

The mesh coordinates of each ridge in each frame were centered around the origin by subtracting from the coordinates of any tracked point in a ridge, the average of the coordinates across all the points in that ridge. Then, to obtain a single stereotypical ridge shape for every phase, the mesh coordinates of each ridge were averaged across all the ridges in a frame and all the frames in the same phase. Displacement field gradients were calculated for each triangle in the mesh,

- between two consecutive phases for the flat plate trials
- between the initial slipping phase and every other phase for the feature plates.

Green-Lagrange strains were estimated from the displacement gradient by the equations:

$$\varepsilon_{xx} = \frac{\partial u}{\partial x} + 0.5 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right] \quad (1)$$

$$\varepsilon_{yy} = \frac{\partial v}{\partial y} + 0.5 \left[\left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] \quad (2)$$

$$\varepsilon_{xy} = 0.5 \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) + 0.5 \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} \right) \quad (3)$$

where ε_{xx} and ε_{yy} are the horizontal and vertical strain, ε_{xy} is the shear strain, and u and v are the displacements along x and y axes, respectively.

The principal components of the strain, e_1 and e_2 , were obtained by eigenvalue decomposition of the strain matrix ε , as reported in (Delhaye et al., 2016 [DOI](#)). The principal strain decomposition consists of a rotation of the reference coordinates so that the shear strain is canceled and the axial strains take their maximal and minimal value. Thus, the principal components e_1 and e_2 represent the maximum tensile and compressive deformation along perpendicular axes.

From the principal strain components, we computed the variation in the area of every triangle composing the mesh as:

$$e_a = \frac{e_1 + e_2}{2} \quad (4)$$

As a measure of overall deformation during sliding of the flat surface, we calculate the maximum shear strain as:

$$e_s = \frac{e_1 - e_2}{2} \quad (5)$$

Statistics

Statistical tests were run in Python using the `scipy.stats` package. As distributions were skewed, we used non-parametric analyses throughout the study. Bonferroni corrections were used when multiple comparisons were made.

Figure supplements

Figure 3—figure supplement 1.

Ridge deformations for all participants during sliding of the lat plate.

Ridge deformation across the four phases for all participants. Shown are the ridge outline for the previous phase (grey shaded regions) with the ridge mesh for the current phase superimposed. Arrows indicate the principal compressive axis for each facet, with colors indicating different orientations relative to vertical. Same style as [Figure 3D](#).

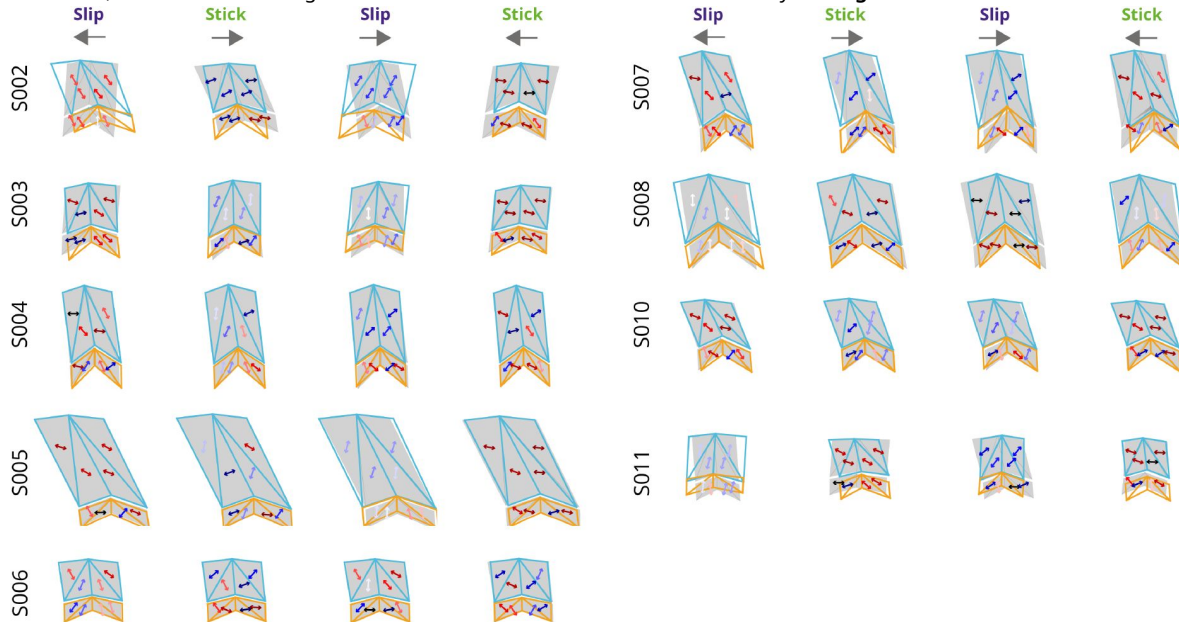
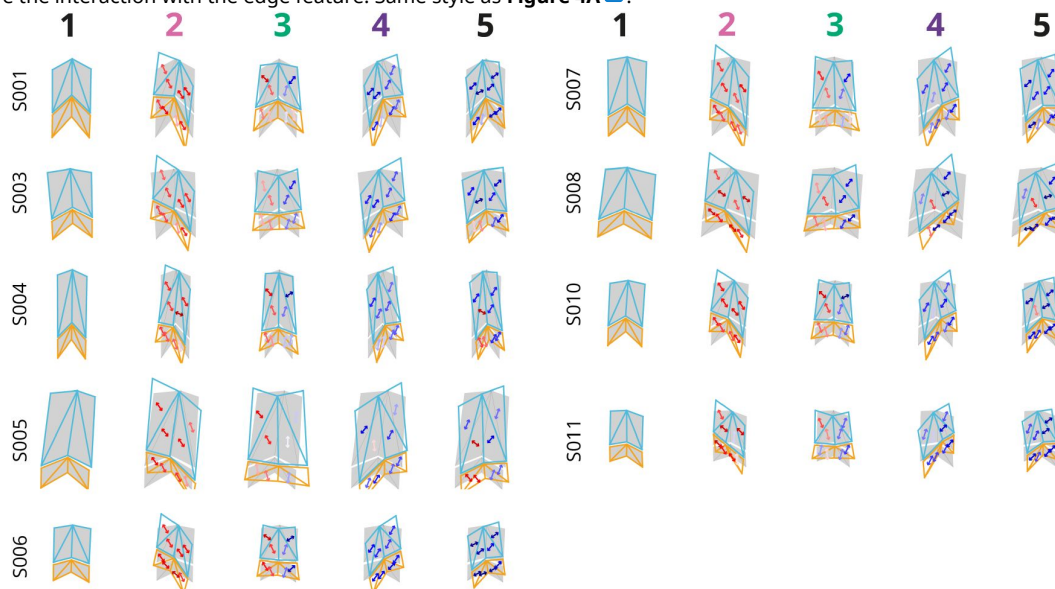


Figure 4—figure supplement 1.

Ridge deformations for all participants during transit of the edge feature.

Ridge deformation and associated principal compressive strain orientation for all participants. Note that all strains are calculated with respect to the mesh shown in the left-most column (phase 1), which represents the ridge during full slip before the interaction with the edge feature. Same style as [Figure 4A](#).



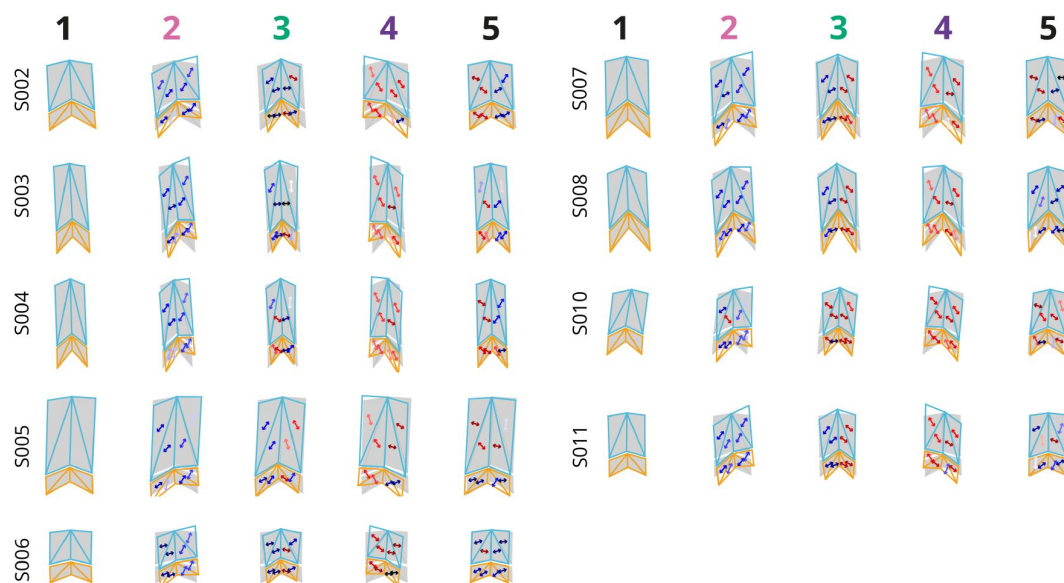


Figure 4—figure supplement 2.

Ridge deformations for all participants during transit of the grooved feature.

Ridge deformation and associated principal compressive strain orientation for all participants. Note that all strains are calculated with respect to the mesh shown in the left-most column (phase 1), which represents the ridge during full slip before the interaction with the edge feature. Same style as **Figure 4D** [↗](#).

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Additional files

Figure 1—video 1 [↗](#)

Additional information

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Reviewer #2 (Public review):

Summary:

The authors investigate sub-skin surface deformations to a number of different, relevant tactile stimuli, including pressure and moving stimuli. The results demonstrate and quantify the tension and compression applied from these types of touch to fingerprint ridges, where pressure flattens the ridges. Their study further revealed that on lateral movement, prominent vertical shearing occurred in ridge deformation, with somewhat inconsistent horizontal shear. This also shows how much the deeper skin layers are deformed in touch, meaning the activation of all cutaneous mechanoreceptors, as well as the possibility of other deeper non-cutaneous mechanoreceptors.

Strengths:

The paper has many strengths. As well as being impactful scientifically, the methods are sound and innovative, producing interesting and detailed results. The results reveal the intricate workings of the skin layers to pressure touch, as well as sliding touch over different conditions. This makes it applicable to many touch situations and provides insights into the differential movements of the skin, and thus the encoding of touch in regards to the function of fingerprints. The work is very clearly written and presented, including how their work relates to the literature and previous hypotheses about the function of fingerprint ridges. The figures are very well-presented and show individual and group data well. The additional supplementary information is informative and the video of the skin tracking demonstrates the experiments well.

Weaknesses:

There are very few weaknesses with the work; rather the authors detail well the limitations in the discussion. Therefore, this opens up lots of possibilities for future work.

Impact/significance:

Overall, the work will likely have a large impact on our understanding of the mechanics of the skin. The detail shown in the study goes beyond current understanding, to add profound insights into how the skin actually deforms and moves on contact and sliding over a surface, respectively. The method could be potentially applied in many other different settings (e.g. to investigate more complex textures, how skin deformation changes with factors like dryness and aging). This fundamental piece of work could therefore be applied to understand skin changes and how these impact on touch perception. It can further be applied to understand skin mechanoreceptor function better and model these. Finally, the importance of fingertip ridges is well-detailed, demonstrating how these play a role in directly shaping our touch perception and how they can shape the interactions we have with surfaces.

<https://doi.org/10.7554/eLife.93554.2.sa2>

Reviewer #3 (Public review):

Summary:

The publication presents unique in-vivo images of the upper layer of the epidermis of glabrous skin when a flat object compresses or slides on the fingertip. The images are captured using OCT and show the strain that fingerprints experience during mechanical stimulation.

The most important finding is, in my opinion, that fingerprints undergo pure compression/tension without horizontal shear, suggesting that the shear stress caused by tangential load is transferred to the deeper tissues and ultimately to the mechanoreceptors (SA-I / RA-I).

Strengths:

Fascinating new insights into the mechanics of glabrous skin. To the best of my knowledge, this is the first experimental evidence of the mechanical deformation of fingerprints when subjected to dynamic mechanical stimulation. The OCT measurement allows unprecedented measurement of skin depth, whereas previous works were limited to tracking surface deformation.

The robust data analysis reveals the continuum mechanics underlying the deformation of the fingerprint ridges.

Weaknesses:

I do not see any major weaknesses. The work is mainly experimental and is rigorously executed.

<https://doi.org/10.7554/eLife.93554.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

This manuscript uses optical coherence tomography (OCT) to visualize tissue microstructures about 1-2 mm under the finger pad skin surface. Their geometric

features are tracked and used to generate tissue strains upon skin surface indentation by a series of transparent stimuli both normal and tangential to the surface. Then movements of the stratum corneum and the upper portion of the viable epidermis are evaluated. Based upon this data, across a number of participants and ridges, around 300 in total, the findings report upon particular movements of these tissue microstructures in various loading states. A better understanding of the mechanics of the skin microstructures is important to understand how surface forces propagate toward the locations of mechanoreceptive end organs, which lie near the edge of the epidermis and dermis, from which tactile responses of at least two peripheral afferents originate. Indeed, the microstructures of the skin are likely to be important in shaping how neural afferents respond and enhance their sensitivity, receptive field characteristics, etc.

Strengths:

The use of OCT in the context of analyzing the movements of skin microstructures is novel. Also novel and powerful is the use of distinct loading cases, e.g., normal, tangential, and stimulus features, e.g., edges, and curves. I am unaware of other empirical visualization studies of this sort. They are state-of-the-art in this field.

Moreover, in addition to the empirical imaging observations, strain vectors in the tissues are calculated over time.

Weaknesses:

The interpretation of the results and their framing relative to the overall hypotheses/questions and prior works could be articulated more clearly. In particular, the major findings of the manuscript are in newly describing a central concept regarding "ridge flanks," but such structures are neither anatomically nor mechanistically defined in a clear fashion. For example, "... it appears that the primary components of ridge deformation and, potentially, neural responses are deformations of the ridge flanks and their relative movement, rather than overall bending of the ridges themselves." From an anatomical perspective, I think what the authors mean by "ridge flanks" is a differential in strain from one lateral side of a papillary ridge to the other. But is it unclear what about the continuous layers of tissue would cause such behaviors. Perhaps a sweat duct or some other structure (not visible to OCT) would subdivide the "flanks" of a papillary ridge somehow? If not due to particular anatomy, then is the importance of the "ridge flank" due to a mechanistic phenomenon of some sort? Given that the findings of the manuscript center upon the introduction of this new concept, I think a greater effort should be made to define what exactly are the "ridge flanks." It is clear from the results, especially the sliding case, that there is something important that the manuscript is getting at with this concept.

We apologize for the confusion around our use of 'ridge flanks'. To recap the overall goal briefly, we wanted to measure the deformation of papillary ridges and their associated sub-surface structures to different tactile stimuli. Capturing these deformations and comparing them against different proposed ideas, for example bending (horizontal shear) of the entire ridge versus differential deformations of different sub-parts, constrains neural activation mechanisms, has implications for how well tactile stimuli can be spatially resolved on the skin, and for whether sub-surface deformations can be easily predicted from surface movements alone. Our mesh was dense enough to compare the stratum corneum and the viable epidermis directly, where we expected some differences due to their previously documented mechanical differences, as well as the ridge flanks, which refers to the two (proximal and distal) sides of a single papillary ridge and their associated structure in the SC and VE (as correctly surmised by the reviewer). Differential behaviour across ridge flanks might be seen, because various observations of the surface of the stratum corneum had suggested mechanical differences between the papillary ridges and the grooves dividing

them, potentially leading to differential deformations of these two halves depending on which direction they were facing tissue with different mechanical properties.

We now provide a clearer definition of ridge flanks in Figure 1 and in the main text. Importantly, existing prior research is better connected to our own investigation in the Introduction and we now specifically explain why we investigate ridge flanks.

The OCT used herein cannot visualize deep and fully into what the manuscript refers to as a "ridge" (note others have previously broken apart this concept apart into "papillary", "intermediate" and "limiting" ridges) near locations of the mechanoreceptive end organs lie at the epidermal-dermal border. Therefore, the OCT must make inferences about the movements of these deeper tissues, but cannot see them directly, and it is the movements of these deeper tissues that are likely driving the intricacies of neural firing. Note the word "ridge" is used often in the manuscript's abstract, introduction, and discussion but the definition in Fig. 1 and elsewhere differs in important ways from prior works of Cauna (expert in anatomy). Therefore, the manuscript should clarify if "ridge" refers to the papillary ridge (visible at the exterior of the skin), intermediate ridge (defined by Cauna as what the authors refer to as the primary ridge), and limiting ridge (defined by Cauna as what the authors refer to as the secondary ridge). What the authors really mean (I think) is some combination of the papillary and intermediate ridge structures, but not the full intermediate ridge. The manuscript acknowledges this in the "Limitations and future work" section, stating that these ridges cannot be resolved. This is important because the manuscript is oriented toward tracking this structure. It sets up the narrative and hypotheses to evaluate the prior works of Cauna, Gerling, Swensson, and others who all directly addressed the movement of this anatomical feature which is key to understanding ultimately how stresses at these locations might move the peripheral end organs (i.e., Merkel cells, Meissner corpuscles).

Thank you for these observations. Indeed, our terminology was not consistent. We have now switched to Cauna's terminology and added additional labels in Figure 1, explaining all mentioned structures in the main text. We have also changed the language in many instances in the main text to make it clearer whether we are referring to individual anatomical ridges (papillary, limiting, etc.) or the whole structure. Additionally, it is now clearer from the start which features are tracked, and we specifically state that intermediate ridges are excluded from our tracking.

Regarding the intermediate ridge, it indeed plays a big role in Cauna's lever hypothesis. Given the intermediate ridge is excluded from our analysis, we can neither prove nor disprove this hypothesis in our current work. However, there are many mechanical mysteries to solve regarding the structures directly above, which are the main focus of this paper. We have rewritten the introduction to make these questions clearer. For example, Cauna observed pliability of the papillary ridges in surface experiments. Swensson found differential expression patterns of keratin in epidermis tissue in and above the intermediate ridges, but the direct mechanical consequences that are proposed in their paper concern the behaviour of papillary ridges, rather than relying on a mechanical role of intermediate ridges. Even Cauna's lever idea implies specific deformation of the stratum corneum, which would be measurable in our study, as the upper handle of the 'lever' needs turning. We observed little movement in accordance with this idea, putting the lever mechanism into question. While this does not rule out a mechanical role of the intermediate ridge, these findings constrain its potential mechanisms.

Reviewer #2 (Public Review):

Summary:

The authors investigate sub-skin surface deformations to a number of different, relevant tactile stimuli, including pressure and moving stimuli. The results demonstrate and quantify the tension and compression applied from these types of touch to fingerprint ridges, where pressure flattens the ridges. Their study further revealed that on lateral movement, prominent vertical shearing occurred in ridge deformation, with somewhat inconsistent horizontal shear. This also shows how much the deeper skin layers are deformed in touch, meaning the activation of all cutaneous mechanoreceptors, as well as the possibility of other deeper non-cutaneous mechanoreceptors.

Strengths:

The paper has many strengths. As well as being impactful scientifically, the methods are sound and innovative, producing interesting and detailed results. The results reveal the intricate workings of the skin layers to pressure touch, as well as sliding touch over different conditions. This makes it applicable to many touch situations and provides insights into the differential movements of the skin, and thus the encoding of touch in regards to the function of fingerprints. The work is very clearly written and presented, including how their work relates to the literature and previous hypotheses about the function of fingerprint ridges. The figures are very well-presented and show individual and group data well. The additional supplementary information is informative and the video of the skin tracking demonstrates the experiments well.

Weaknesses:

There are very few weaknesses in the work, rather the authors detail well the limitations in the discussion. Therefore, this opens up lots of possibilities for future work.

We thank the reviewer for these encouraging comments.

Impact/significance:

Overall, the work will likely have a large impact on our understanding of the mechanics of the skin. The detail shown in the study goes beyond current understanding, to add profound insights into how the skin actually deforms and moves on contact and sliding over a surface, respectively. The method could be potentially applied in many other different settings (e.g. to investigate more complex textures, and how skin deformation changes with factors like dryness and aging). This fundamental piece of work could therefore be applied to understand skin changes and how these impact touch perception. It can further be applied to understand skin mechanoreceptor function better and model these. Finally, the importance of fingertip ridges is well-detailed, demonstrating how these play a role in directly shaping our touch perception and how they can shape the interactions we have with surfaces.

Reviewer #3 (Public Review):

Summary:

The publication presents unique in-vivo images of the upper layer of the epidermis of the glabrous skin when a flat object compresses or slides on the fingertip. The images are captured using OCT, and are the process of recovering the strain that fingerprints experience during the mechanical stimulation.

The most important finding is, in my opinion, that fingerprints undergo pure compression/tension without horizontal shear, hinting at the fact that the shear stress caused by the tangential load is transferred to the deeper tissues and ultimately to the mechanoreceptors (SA-I / RA-I).

Strengths:

Fascinating new insights into the mechanics of glabrous skin. To the best of my knowledge, this is the first experimental evidence of the mechanical deformation of fingerprints when subjected to dynamic mechanical stimulation. The OCT measurement allows an unprecedented measurement of the depth of the skin whereas previous works were limited to tracking the surface deformation. - The robust data analysis reveals the continuum mechanics underlying the deformation of the fingerprint ridges.

Weaknesses:

I do not see any major weaknesses. The work is mainly experimental and is rigorously executed. Two points pique my curiosity, however:

(1) How do the results presented in this study compare with previous finite element analysis? I am curious to know if the claim that the horizontal shear strain is transferred to the previous layer is also captured by these models. The reason is that the FEA models typically use homogeneous materials and whether or not the behavior in-silico and in-vivo matches would offer an idea of the nature of the stratum corneum.

Very few modeling studies have examined combined normal and tangential loading of the fingertip. Additionally, results are often expressed in terms of Von Mises stresses, and not deformation [1,2], making direct comparison challenging. Nevertheless, one multilayered study [3] supports our finding that the largest deformations are found in deeper tissues.

(1) Shao, F., Childs, T. H. C., Barnes, C. J. & Henson, B. Finite element simulations of static and sliding contact between a human fingertip and textured surfaces. *Tribology International* 43, 2308–2316 (2010).

(2) Tang, W. et al. Investigation of mechanical responses to the tactile perception of surfaces with different textures using the finite element method. *Advances in Mechanical Engineering* 8, (2016).

(3) Amaied, E., Vargiolu, R., Bergheau, J. M. & Zahouani, H. Aging effect on tactile perception: Experimental and modelling studies. *Wear* 332–333, 715–724 (2015).

(2) Was there a specific reason why the authors chose to track only one fingerprint? From the method section, it seems that nothing would have prevented tracking a denser point cloud and reconstructing the stain on a section of the skin rather than just one ridge. With such data, the author could extend their analysis to multiple ridges interaction and get a better sense of the behavior of the entire strip of skin.

We apologise for the confusion regarding this point. While in our illustration and the accompanying videos, we only show a single tracked ridge for clarity, we do indeed track all visible ridges in every frame. As imaging slices were 4 mm wide, often 8-9 ridges were visible concurrently. However, during the sliding experiments the skin was sometimes dragged along with the stimulus, causing some ridges to disappear from view for certain periods and then re-enter the frame. This would make it difficult to expand the analysis to multiple ridges, but in any case, we found neighbouring ridges to behave very consistently within a given trial, so that their mechanical behaviour (relative to the tactile feature, if any) could be averaged in the analysis.

Reviewer #1 (Recommendations For The Authors):

Discussion, line 213, "Thus, the primary mechanism through which the ridge conforms to the object involves the relative movement and shearing of the ridge flanks, rather than

relying on the grooves as articulated joints." I don't see this as definitely proven in the imaging and analysis. This could be a hypothesis to come from this work for further evaluation but is a quite strong statement not obviously supported by the evidence.

We have rephrased this statement as a proposal for further testing:

"Therefore, we propose that the primary mechanism through which a ridge conforms to an object might involve the relative movement and shearing of the ridge flanks, rather than relying on the grooves as articulated joints."

Discussion, line 220, "Our findings strongly indicate that the majority of the surface movement of the skin was observed by deeper tissue rather than surface layers of the skin." But since there are no measurements of such tissues, or of collagen bundle tightening, etc. it is not obvious to me how this can be proven as it is not directly observable and was not modeled.

We have reworded this paragraph to be more cautious and have included potential avenues for future testing of this idea:

"It is possible that the majority of the surface movement of the skin was absorbed by deeper tissues rather than the surface layers of the skin imaged in the present study. If that is the case, recent modeling work has suggested that tissue deformations are highly dependent on the orientation of collagen fibers in these tissues (Duprez et al., 2024), which might be amenable to tracking in future OCT work to test this idea directly. Additionally, previous work investigating tactile afferent responses to tangential skin movements has reported strong activation of SA-2 receptors, thought to measure skin stretch mainly in deeper tissues (Saal et al., 2025), providing further indirect evidence."

Figure 1, A. As noted elsewhere, there are issues with the naming of the anatomy, and there is no definition of the concept of "ridge flanks." Also, it does not indicate the depth point to which OCT can resolve.

We have updated and expanded the labels in Figure 1A to clarify the anatomy (along with changes in the text described above). Figure 1C now includes a sentence about the resolvability of features below the mesh:

"Detail view of a single OCT frame showing ridged skin structure and clear boundary between the stratum corneum and viable epidermis. A mesh covering the stratum corneum and the upper part of the viable epidermis (without the intermediate ridge) is overlaid spanning a single papillary ridge. The border between the viable epidermis and dermis is less clearly delineated, but some deeper features are resolved less well."

The concept of a ridge flank is now illustrated in Figure 1B(i) and Figure 1B(iv), and referred to in both the caption and main text. Updated figure caption text:

"These deformations need not apply to the whole ridge structure but might affect different parts separately, e.g. via shearing in different directions across both ridge flanks as shown on the far right

(see darker shading to highlight a single ridge flank)."

Updated text in the main manuscript:

"Additionally, if there are indeed mechanical differences between papillary ridges and their neighbouring grooves at the level of the stratum corneum, this might result in differential movements of the two sides of each papillary ridge, here referred to as ridge flanks (see Figure 1B-iv, right, for a potential example)."

Note that Figure 4B also includes an illustration of this concept.

Figure 1, B. This mechanical representation does not capture the entirety of the papillary-intermediate ridge unit in question, as set up by the authors in the introduction. Also, in the caption it is not ridge deformation, but upper SC and VE deformation. And the OCT cannot resolve the whole ridge.

We have reworded the figure caption”

“Potential deformations of the tracked ridge structure, including the stratum corneum and the bulk of the viable epidermis, during tactile interactions, with arrows indicating the directions of relative deformation. [...]”

Importantly, the main manuscript text has been rewritten in the introduction section to clarify our research question and how much of the sub-surface ridge structure is tracked:

“From a mechanical standpoint, these conflicting interpretations raise the question of how the outermost two skin layers typically deform at the resolution of single papillary ridges, whether by tension, compression, or shear (see examples in Figure 1B). Additionally, such deformations might apply to individual papillary ridges and all their sub-surface structures equally, for example horizontal shearing that bends the papillary ridge in a certain direction, while levering its sub-surface aspects in the opposite direction. Conversely, individual parts of the ridge structure might deform differently. For example, the viable epidermis might deform to a different extent or in different directions due to its lower stiffness and different morphology. Additionally, if there are indeed mechanical differences between papillary ridges and their neighbouring grooves at the level of the stratum corneum, this might result in differential movements of the two sides of each papillary ridge, here referred to as ridge flanks (see Figure 1B-iv, right, for a potential example). To empirically address these questions, we employed Optical Coherence Tomography (OCT) to precisely measure the sub-surface deformation of individual fingerprint ridges in response to a variety of mechanical events. Specifically, we focused on the stratum corneum and the bulk of the viable epidermis (excluding intermediate ridges), which could be robustly resolved and tracked by our setup.”

Figure 1, C: While it is noted in the caption that the locations of the intermediate and limiting ridges, as well as the collagen bundles, are clearly visible, it is not clear to me, although the caption uses these words. This is especially the case below the orange mesh. From the picture, and because this is not labeled, it leaves it up to my interpretation, it seems like the secondary ridge (limiting) is larger than the primary (intermediate).

We have reworded the caption as follows:

“Detail view of a single OCT frame showing ridged skin structure and clear boundary between the stratum corneum and viable epidermis. A mesh covering the stratum corneum and the upper part of the viable epidermis (without the intermediate ridge) is overlaid spanning a single papillary ridge. The border between the viable epidermis and dermis is less clearly delineated.”

Indeed, while the intermediate ridge was often visible in the OCT images, its size was rather inconsistent and it could appear as larger or smaller than the limiting ridge, while in histological images it is generally shown as larger (however note that there is somewhat limited data). This difference might be due to imaging artifacts, e.g. limited visibility into the deeper tissues, might reflect individual differences between participants, or could indicate that intermediate ridges are not of a consistent height in the (out-of-plane) direction along a given ridge. We have clarified this in the Limitations section of the Discussion:

“[...] while we could confidently track landmarks associated with the stratum corneum, we could not reliably identify intermediate ridges in the viable epidermis, though they were visible in some of the frames, limiting the depth of the fitted mesh. We hypothesize that the additional depth of these ridges combined with their slender morphology might have degraded the signal. 3D OCT imaging (see below) might help to resolve these features in future work and settle open questions regarding their precise morphology.”

Figure 1, D, and E: How do these measurements compare with the literature? They seem reasonable to me based on a cursory review, but there is a need to directly compare, especially since measurements in this context with the OCT are novel and could be valuable.

We have clarified this in the main text and added more references to the existing literature:

“We measured an average ridge width of 0.47 mm across participants (Figure 1D), consistent with previous studies (Moore, 1989; Ohler and Cummins, 1942). Average skin layer thickness was 0.38 mm for the stratum corneum and 0.12 mm for the viable epidermis across our dataset (Figure 1E), again in agreement with previous studies using both in vivo imaging and ex vivo histology (Fruhstorfer et al., 2000; Lintzeri et al., 2022; Maiti et al., 2020).”

Abstract 4th sentence's structure makes me think that hundreds of individual fingerprint ridges can be tracked at the same time. Perhaps it could be tweaked to clearly indicate that hundreds were tracked between trials between participants.

We have changed the sentence to now read:

“Here, we used optical coherence tomography to image and track sub-surface deformations of hundreds of individual fingerprint ridges across ten participants and four individual contact events at high spatial resolution in vivo.”

Introduction, 1st sentence, the fingertip per se is not an organ, though the skin is an organ.

Changed the wording from “organ” to “structure”.

Introduction, 1st sentence, "... that convert skin deformations ..." Need to add word skin to be clear.

Done.

Introduction, 3rd paragraph, "Alternately, the grooves may be stiffer or less ...". In this paragraph, and this sentence in particular, Cauna is cited and the words grooves and ridges are used. But this is not adequately explained. Cauna had distinct terminology, where he referred to papillary, intermediate, and limiting ridges, that exist in addition to ready ridges. It is important because the manuscript uses the word "ridges" in a non-specific way. This is done not just here but throughout the manuscript, and is central to the questions which can be addressed with OCT.

Anatomy has been better defined and more extensively labelled in Figure 1A, including labels for ‘papillary ridges’ and ‘grooves’. We have reworded this paragraph to better explain the concepts and how they relate to the subsequent analyses in the paper

“Consequently, the mechanical response of the skin below its immediate surface remains largely unknown, leading to conflicting interpretations in the literature. For instance, it has

been proposed that the papillary ridges are stiffer than the neighbouring grooves (Swensson et al., 1998), which might imply that normal loading of the skin might not affect the ridges' profile appreciably. Conversely, other observations have suggested that the grooves are relatively stiff, allowing the papillary ridges to deform considerably (Cauna, 1954; Johansson and LaMotte, 1983). However, the sub-surface consequences of this putative pliability during object contact or stick-to-slip transitions (see e.g. Delhay et al., 2016) are unclear: the whole ridge structure might bend as proposed in Cauna's lever mechanism (Cauna, 1954), but this view has proved controversial (see e.g. Gerling and Thomas, 2008), with direct empirical evidence lacking."

Figure 1. Avoid red-green dots for colorblind accessibility. PMMA is not in the caption.

We have switched the colors of the mechanoreceptors in panel A to a colorblind-friendly scheme. We now also specify the material of the plates in the figure 1 caption.

Results, line 102. "... papillary ridge structure...." Is this the ridge to which is being referred?

In conjunction with the updated labeling in Figure 1A, we have updated the terminology throughout the paper to be more consistent.

Results, line 99. "We noted a small increase in the area of the stratum corneum, which was likely an artifact due to the fit of the mesh to the ridge's curvature ..." There is very little discussion of Fig. F's finding related to an increase in area in the SC and decrease in the VE. It makes me question if this finding in this panel is an artifact. With stiff tissue like stratum corneum, how would the area increase?

This finding could be a measurement artifact or it could be the result of skin from neighbouring regions pushing into the imaged space. We have reworded the brief description in the Results:

"We noted a small increase in the area of the stratum corneum, which was possibly an artifact due to the imperfect fit of the mesh to the ridge's curvature (but see Discussion for an alternative explanation)."

Additionally, we have added a short section in the Discussion in the Limitations section:

"Some of our tactile interactions might have caused skin deformations out-of-plane that were thus not measurable. For example, the slight increase in thickness of the stratum corneum under normal load might be explained as a measurement artifact due to the coarse nature of the mesh fitted, but could alternatively reflect tissue from out-of-plane regions pushing into the imaged space. Indeed, recent surface measurements of the skin's behaviour during initial object contact have reported compression of the skin in the plane parallel to its surface (Dumont et al., 2025), which would result in increasing thickness, assuming that the stratum corneum is incompressible. Future studies could consider creating three-dimensional reconstructions of the fingerprint structure to study such effects."

Figure 3. The colors used in slip and stick are not colorblind accessible.

We have changed the background colors in Figure 3A,B,C to a colorblind accessible version.

Results, line 151, "Thus, most of this shearing must be sustained by deeper tissues." But there are no direct observations as such. Also, in the next sentence, "collagen fiber bundles" are referred to in a non-specific way. This section is highly speculative with no

systematic visualization of these structures, and should probably be moved to the discussion.

We have reworded this sentence to be more cautious. We have now also highlighted collagen fiber bundles visible in the figure. Systematic analysis of these is beyond the scope of the present study, as these were not tracked, but might be possible in future studies. The reworded sentence reads as follows:

“Thus, it is possible that shearing is sustained by deeper tissues, an effect that could be tested in future studies by directly tracking the angle and orientation of collagen fiber bundles anchoring the epidermis to deeper tissues (see highlighted examples in Figure 3B).”

Results, line 161, "Horizontal shear ..." do you mean surface shear, per the Fig. 1 definition?

For consistency, we have changed the labels to ‘Horizontal shear’ and ‘Vertical shear’ in Figure 1A(iii) and Figure 1A(iv) as these are the terms used throughout the paper.

Discussion, line 198, "... flatten even at relatively low forces." This is an interesting point and it would be useful to note how low exactly.

We have reworded this sentence to better reflect the findings described earlier:

“We found that individual ridges tended to flatten considerably at relatively low forces of 0.5 N, with higher forces increasing deformations only moderately.”

Reviewer #2 (Recommendations For The Authors):

Minor comments that could improve the paper even further

In the abstract, it may be good to specify that the stimuli were all applied to the finger, this was not an active, self-generated tactile interaction, e.g. change 'in response to a variety of tactile stimuli' to 'in response to a variety of passively-applied tactile stimuli'.

Done.

Comment on the grey/blue colours in the figures. I like the combination of blue/orange for different conditions, but sometimes the blue is very difficult to see against the grey background. Is there any way of making the grey background shading lighter and/or the blue darker/more vivid?

We have changed the color of the SC mesh to a darker shade of blue, which is more easily distinguished from the grey background. This applies to figures 2B/C, 3D, 4A/B/D/E, and all supplementary figures.

Methods. Could you please add a little more detail about exactly where the images were taken, e.g. in the exact middle of the fingerpad, at the fingertip? Did you line up the skin fingerprint ridges to be in a plane? It is just to better understand how the stimulus moved against the skin, which itself is rounded, and whether it was at a point where the ridges were relatively linear or curved.

We have added the following text in the “Experimental set-up” section of the Methods:

“The participant's finger was secured in a finger holder, which was positioned in such a way that the flat part of the fingertip distal to the whorl made initial contact with the plate as it was lowered onto the fingertip. The scanner was positioned such that its scan path aligned

with the distal-proximal axis of the plate, targeting the centre line of the fingerpad so that the fingerprint ridges were oriented orthogonally to the line scan.”

and

“For these experiments, imaging focused on the central flat part of the contact area, such that all fingerprint ridges visible in the imaged region were in contact with the plate throughout the trial.”

Methods. There is no section about statistics, yet you do use them in the paper. It may be good to add a few details in the methods to outline the package you used to do the statistics, as well as why you chose the tests you carried out.

We have added a new Statistics section at the end of the Methods:

“Statistical tests were run in Python using the scipy.stats package. As distributions were skewed, we used non-parametric analyses throughout the study. Bonferroni corrections were used when multiple comparisons were made.”

A very minor point. Discussion, line 210: 'In this study...' is vague, which study exactly? It is preferable to be more precise, e.g. 'In the present/current study...'.

Fixed.

Discussion. One point you may want to add is the possibility of looking at other skin regions. For example, would this approach work on the palm, on border glabrous/hairy skin, on various hairy skin sites, and on the foot? The possibilities could be endless if it could be applied anywhere, but it may depend on the technical positioning and skin itself. However, it would be interesting to know.

We have added the following text at the end of the Discussion section:

“Finally, while we focused on the fingertip only, many other skin regions present interesting mechanical challenges waiting to be explored. The general ridged structure observed on the fingertip is common to all glabrous skin, but the local ridge mechanics might still differ: glabrous skin on the foot sole exhibits some morphological differences in order to support large weights that might well influence its mechanical response (Boyle et al., 2019). For example, the morphology of transverse ridges (running orthogonal to and connecting limiting with intermediate ridges) differs across regions on the foot sole (Nagashima and Tsuchida, 2011) and very likely from the hand (Yamada et al., 1996). Our method should be directly applicable to study deformations of these ridges, though three-dimensional observations might be needed to resolve some of the open questions. Hairy skin in contrast differs from glabrous skin in that the stratum corneum is much thinner. It also lacks the clearly organised ridge structure, but exhibits more loosely oriented skin folds instead, which very likely also serve a mechanical function (Leyva-Mendivil et al., 2015) and in principle are amenable to study using OCT.”

In the last lines of the discussion, you mention the possible effects of skin moisturization. The Tomlinson et al. paper refers to the hydration of the skin with regard to water, which I would say is a slightly different factor. I think you can mention this paper and talk about the water level of the skin/hydration, but also add specifically that moisturization (i.e. by an emollient, humectant, or occlusive substance) is another factor to consider (e.g. effects found by Dione et al, 2023 Sci Rep). Overall, these two points relate to the dryness of the skin and the humidity of surfaces being contacted, therefore you could expand on both.

Thank you for the correction! We now mention both skin hydration and moisturization separately in this section.

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